David Silva

SENIOR SOFTWARE ENGINEER · AI ENGINEERING

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PROFESSIONAL SUMMARY

Senior software engineer with 15+ years running production systems at GOV.UK, the Ministry of Justice, Indeed and in fintech. At Tembo I lead how the mortgage team builds with AI. Agents write most of the code. I set the specs, the review and the quality gates that decide what ships. I still build across backend and frontend, and I coach engineers to work this way.

KEY SKILLS

AI ENGINEERING	Claude Code, Codex, agent harnesses, MCP, AI code review, evals, AWS Bedrock
LANGUAGES & FRAMEWORKS	Ruby/Rails, TypeScript/Next.js, JavaScript, SQL, Python, Rust
ARCHITECTURE	Distributed systems, event-driven, modular monoliths / Rails engines, microservices
DATA & MESSAGING	PostgreSQL, MySQL, Redis, Cassandra, Sidekiq, Kafka, Elasticsearch, Snowflake
CLOUD & OPERATIONS	AWS (Lambda, RDS, Bedrock), Docker, GitHub Actions, NewRelic, Datadog, Sentry
PRACTICES	TDD/BDD, end-to-end testing, Pair Programming, Technical Mentoring, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Tembo

Feb 2026 – Present

Senior engineer on the mortgage team (four engineers, working directly with the CTO). I own the team's AI engineering practice and ship across backend, frontend and production on an FCA-regulated platform handling ~1.7M requests a day. `Ruby, Rails, TypeScript, Next.js, AWS Bedrock.`

- Built the team's **agentic delivery harness** and work spec-first. I set intent and acceptance criteria, agents plan and build, I validate the output. It covers planning, PR review, triage and releases, and the whole team uses it.
- **Review AI-generated code** for spec deviations, security, and logic that passes the tests but misses the intent. Reviewer agents check migrations, performance and engine boundaries. Recurring faults become lint rules or tests, and a human approves every release.
- Put **quality gates** in CI: custom RuboCop cops for migration safety and engine boundaries, and end-to-end tests that run a full customer journey across the Next.js app and the Rails backend.
- Designed the **evaluation framework** for mortgage recommendations: 100+ synthetic customer profiles recorded against live lender APIs and replayed in CI against a baseline. Regressions name the profile and product that moved, drifted recordings fail as out of date, and brokers sign off expected outcomes.
- Made **AWS Bedrock document extraction** reliable: measured the bank-statement failure rate, traced failures to replies cut off at the output token limit and added a stop-reason check, and pinned temperature to 0 so results repeat. Contributed the Bedrock provider upstream to ActiveAgent.
- Own the **solution architecture for the AI remortgage journey** (self-service, no broker): design proposals, an end-to-end gap analysis, and a production foundation with feature flags and an encrypted append-only decision log as the FCA audit trail.
- Built a **shared knowledge base** indexing business meetings and decisions, 550 curated GitHub issues and the codebase, so engineers and agents start each feature with the right context.
- **Coach the team into AI-native working** through pair programming, a weekly session on new tools, and agreeing harness changes with the team before rollout.
- Built a **unified case-management tool** (22 systems to one, case handling **~65 to ~10 minutes**) and automated **lender submission** end to end (1,100+ fields, 40–60 minutes to **about 4**), taking the business halfway to **15,000 mortgages without adding brokers**.
- Established **production observability from scratch**: six NewRelic dashboards including the board-level OKR view, with alerts feeding agent triage. Cut the **deploy pipeline from ~2 hours to 35 minutes**.

Backend Senior Software Engineer, Indeed — 2022 – 2025

Led backend engineering on Indeed Flex, a hiring and staffing marketplace serving 1M+ workers across US/UK markets. `Ruby, Rails, PostgreSQL, Redis, Kafka, OpenAI.`

- **Technical lead on two briefs:** Fulfilment Efficiencies (wrote the brief, planned delivery, built the backend and shipped it) and Agency Role and Compliance Configurations.
- Led a 6-month redesign of the worker penalty appeals system with product and operations: **support tickets down 80%** and wrongful bans eliminated.
- Led the response to a **critical incident affecting 100K+ workers**, running the data migration that restored service.
- Built one of Indeed's first production **AI features**: a job-posting chat on OpenAI GPT with few-shot prompting, serving thousands of interactions a day.
- Spotted Google's **FCM push deprecation** early, forked the unmaintained Rpush library, planned the migration and handed it to the team, who shipped before the deadline.
- Built tooling that generates API and event-catalog documentation from the code, and removed deprecated endpoints. 333 commits across bookings, worker conduct, agencies and the Flex Agent API.
- **Mentored engineers** through pair programming and 1-on-1s, contributing to promotions, and onboarded new hires.

Senior Software Engineer, Ministry of Justice — 2020 – 2022

Built government services for millions of UK citizens. `Ruby, Rails, PostgreSQL, GOV.UK PaaS.`

- Built a service to apply to court about child arrangements, validated through extensive user research, improving access to justice for families.
- Led development of a service to check when to disclose cautions or convictions, translating complex legal requirements into custom algorithms with policymakers and legal experts.
- **Transformed court payment processing** with an online system that eliminated weeks-long delays, removed manual cheque handling, and enabled instant fund allocation to court budgets.

Senior Software Engineer, Creditspring — 2018 – 2020

Joined a pre-launch fintech startup and architected scalable infrastructure, growing from 0 to 100K+ paying customers (500K+ registered users) while maintaining reliability.

- Architected **event-driven systems** using Rails, PostgreSQL, Redis, and Sidekiq, ensuring complete financial auditability and scalability for affordable borrowing products.
- Established monitoring and reliability practices maintaining **99.9%+ uptime** during rapid customer growth.

Software Engineer, Government Digital Service — 2016 – 2018

Developed core GOV.UK features for 60M+ monthly visitors. `Ruby, Rails, Elasticsearch.`

- Led development of GOV.UK **Step by Step pages**, simplifying complex government processes into clear user journeys for millions of citizens.
- Enhanced GOV.UK **search with Elasticsearch**, optimising performance and accuracy for millions of daily searches and phasing out legacy systems.

EARLIER EXPERIENCE

Software Engineer roles at Unii, Sticky9, HouseTrip and Think Orange. Detail on LinkedIn.

PROJECTS & PUBLICATIONS

Modular Rails: Architecture for the Long Game — technical book on building modular monolith applications with Rails engines as a principled alternative to microservices; well received in the Ruby on Rails community.

Building Your Own Roguelike: A Practical Guide — technical book on game development and event-driven architecture in Ruby, with an accompanying open-source game.